GROUP 7 | SMART CAMPUS

Energy Dashboard Developer Handover

MQTT connection, Android IoT MQTT Panel setup, widget topics and integration guide

Purpose
Give the dashboard developer the exact broker settings, command topics, telemetry topics, panel configurations and sample layout for visualising the Energy Saving subsystem.

BROKER • broker.hivemq.com:1883 (TCP)

CONTROLLER • Arduino Uno WiFi Rev2

APP • IoT MQTT Panel (Android)

HANDOVER STATUS

Working integration: The Energy v3 Arduino sketch has been tested with HiveMQ. It accepts phone commands for operating mode, environmental scenario and crowd level, then publishes actual/effective readings, lighting, occupancy, HVAC and DHT11 status under the sitg7/room1/energy/ topic branch.

This guide covers broker setup, Android IoT MQTT Panel configuration, command controls, output topics, sample layout, test scenarios and final integration cautions. The working Arduino sketch is shared separately.

SAMPLE DASHBOARD LAYOUT

SMART CAMPUS
GROUP 7 | ENERGY MANAGEMENT

ONLINE
MQTT connection

01 SYSTEM CONTROLS

Operating Mode: LIVE / SIMULATION

Environmental Scenario: NORMAL | HOT | HUMID | HOT_HUMID

Crowd Level: NONE | LOW | MEDIUM | HIGH

ACTUAL TEMPERATURE
30.4 °C
Physical DHT11

ACTUAL HUMIDITY
73.3 %
Physical DHT11

EFFECTIVE TEMPERATURE
33.0 °C
Value used by HVAC

EFFECTIVE HUMIDITY
85.0 %
Value used by HVAC

03 ENERGY STATUS

Occupancy: OCCUPIED
Ambient: DARK
Lighting: ON
Crowd: HIGH

HVAC: HIGH
Detailed HVAC: HIGH

04 DEVICE HEALTH

DHT11: NORMAL
MQTT: CONNECTED

Only display health states actually published by the Arduino. Current energy sketch publishes DHT11 health; PIR/LDR fault topics are not implemented.

Design principle: Keep controls first, actual physical readings separate from effective HVAC inputs, and calculated outputs together. This sample uses illustrative values only; it is a layout guide, not a live screenshot.

1. Connect IoT MQTT Panel to HiveMQ

The dashboard is built in the Android version of IoT MQTT Panel. Create one broker connection first, then add a dashboard under that connection.

Broker connection settings

Field	Value
Connection name	SIT Group 7 (or a clear dashboard-specific name)
Client ID	Use a unique ID, e.g. sitg7-dashboard-01
Broker address	broker.hivemq.com
Port	1883
Network protocol	TCP
Username / Password	Leave both blank
SSL / TLS	OFF
Auto connect	ON
Clean session	ON
Keep alive	300 seconds is acceptable

Client ID rule: Every connected device must use a different Client ID. The Energy Arduino currently uses sitg7-energy-uno02. The dashboard phone/tablet must use its own ID, such as sitg7-dashboard-01. Do not copy the Arduino ID onto the phone.

Create the dashboard

- In IoT MQTT Panel, choose Setup a Connection and enter the settings above.
- Under Add Dashboard, create a dashboard named Smart Campus Energy. It can be set as the connection home.
- Tap Create to save the connection.
- Return to Connections and tap the connection/cloud status icon to connect. The cloud icon should indicate an active connection.
- Open the dashboard and use the + button to add panels.

Network requirements and public broker caution

- The Arduino Uno WiFi Rev2 uses a 2.4 GHz Wi-Fi network. A phone hotspot is acceptable if it offers 2.4 GHz compatibility and internet access.
- The Android phone/tablet and Arduino do not have to use the same Wi-Fi network, but both must be able to reach the public broker.
- HiveMQ public broker is for prototype/testing. Do not send passwords, personal information or sensitive data.
- If the app is closed or the device disconnects, reopen the saved connection and reconnect. Auto connect is enabled but verify the status.

2. Phone-to-Arduino control panels

Create three Combo Box panels. These panels publish commands from the phone to the Arduino. The current Energy v3 sketch subscribes to all three topics under sitg7/room1/cmd/#.

Common Combo Box settings

- Disable dashboard prefix topic: TICK this box for each panel. The Arduino expects the exact topics below.
- Subscribe Topic: Leave blank for these command controls.
- Use icon for option: Leave unticked unless icons are intentionally configured.
- QoS: 0. Retain: OFF. Use the default clean-session connection settings.
- For every option, the Label is the text shown on the phone; the Payload is the exact message sent to the Arduino. Payload spelling matters.

Panel A - Operating Mode

Field	Value
Panel name	Operating Mode
Topic	sitg7/room1/cmd/mode
Option 1 label / payload	Live Sensor / LIVE
Option 2 label / payload	Simulation / SIMULATION

Panel B - Environmental Scenario

Field	Value
Panel name	Environmental Scenario
Topic	sitg7/room1/cmd/scenario
Normal	Label: Normal | Payload: NORMAL
High Temperature	Label: High Temperature | Payload: HOT
High Humidity	Label: High Humidity | Payload: HUMID
Hot & Humid	Label: Hot & Humid | Payload: HOT_HUMID

Panel C - Crowd Level

Field	Value
Panel name	Crowd Level
Topic	sitg7/room1/cmd/crowd
None	Label: None | Payload: 0
Low	Label: Low | Payload: 1
Medium	Label: Medium | Payload: 2
High	Label: High | Payload: 3

Command flow: Phone Combo Box → MQTT command topic → Arduino subscription → update mode/scenario/crowd variable → HVAC logic recalculates. After an Arduino restart, reselect the mode, scenario and crowd level because the current sketch starts with LIVE, NORMAL and crowd 0.

3. Arduino-to-phone display panels

The current Energy v3 sketch publishes telemetry approximately every 2 seconds while MQTT is connected. Add Text Output panels to subscribe to individual topics. For status cards, select the display option that shows the latest message. Keep a separate diagnostic output panel for viewing all recent messages if useful.

Environmental monitoring topics

Panel name	Subscribe topic	Meaning
Actual Temperature	sitg7/room1/energy/actual_temp	Physical DHT11 temperature, °C
Actual Humidity	sitg7/room1/energy/actual_hum	Physical DHT11 relative humidity, %
Effective Temperature	sitg7/room1/energy/temp	Value used by HVAC: live or simulated, °C
Effective Humidity	sitg7/room1/energy/hum	Value used by HVAC: live or simulated, %

Energy status and device state topics

Panel name	Subscribe topic	Payload examples
Crowd Level	sitg7/room1/energy/crowd	0 / 1 / 2 / 3
HVAC Setting	sitg7/room1/energy/hvac	OFF / LOW / MEDIUM / HIGH
HVAC Detailed Setting	sitg7/room1/energy/hvac_detail	OFF / LOW / LOW-MEDIUM / MEDIUM / MEDIUM-HIGH / HIGH
Lighting Status	sitg7/room1/energy/lighting	ON / OFF
Room Occupancy	sitg7/room1/energy/occupancy	OCCUPIED / VACANT
Ambient State	sitg7/room1/energy/ambient	DARK / BRIGHT
Operating Mode	sitg7/room1/energy/mode	LIVE / SIMULATION
Scenario	sitg7/room1/energy/scenario	NORMAL / HOT / HUMID / HOT_HUMID
DHT11 Status	sitg7/room1/energy/dht_status	NORMAL / FAULT

Text Output panel settings

- Panel type: Text Output. (In the Android app used during setup, the available panel was Text Output; do not look for a separate Text Log panel.)
- Disable dashboard prefix topic: TICK. Enter the exact topic in the panel Topic field.
- For single-value status panels, display the latest received message. For a diagnostic panel, use wildcard topic sitg7/room1/energy/# and select all received payloads if available.
- QoS 0; leave Retain OFF unless the group deliberately changes the retained-message design.

Actual vs effective values: Actual temperature/humidity always represent the physical DHT11. Effective temperature/humidity are the values currently used by HVAC. In SIMULATION mode these are the selected preset values, while actual readings remain separately visible.

4. Visualisation and dashboard layout

Recommended layout order: put the controls near the top, then the physical and effective environmental readings, then energy decisions, and finally health/diagnostic information. Keep the dashboard usable on a phone: use paired half-width widgets where the app grid allows it.

Dashboard section	Recommended widgets	Topic / configuration
01 - SYSTEM CONTROLS	3 Combo Boxes: Operating Mode, Environmental Scenario, Crowd Level	Command topics from page 3
02 - ENVIRONMENTAL MONITORING	4 Text Outputs initially; optionally replace effective readings with Gauges	actual_temp, actual_hum, temp, hum
03 - ENERGY STATUS	Text Outputs or Multi-State Indicators for occupancy, lighting, HVAC, ambient	occupancy, lighting, hvac, ambient
04 - DEVICE HEALTH	Text Output or LED/Multi-State Indicator for DHT11 status	dht_status
DIAGNOSTICS	One Text Output with wildcard subscription	sitg7/room1/energy/#

Optional Gauge settings

Gauge	Subscribe topic	Suggested scale	Unit
Effective Temperature	sitg7/room1/energy/temp	0 to 50	°C
Effective Humidity	sitg7/room1/energy/hum	0 to 100	%

Optional line graphs

Graph	Subscribe topic	What it shows
Temperature trend	sitg7/room1/energy/temp	Changes in effective HVAC temperature over time
Humidity trend	sitg7/room1/energy/hum	Changes in effective HVAC humidity over time

- Use the effective topics for graphs if the goal is to show exactly what the HVAC algorithm is using. Keep actual readings in separate Text Outputs or additional graphs if desired.
- Use a multi-state indicator for OCCUPIED/VACANT and ON/OFF if the widget supports mapping the text payloads to states.
- Use a Text Output for HVAC detailed setting because it can show intermediate values LOW-MEDIUM and MEDIUM-HIGH.
- Graph history in the app may be limited by the installed version or app plan. Use Google Sheets/Apps Script for durable historical records rather than assuming the dashboard stores all past messages.

Important display limitation: The current Arduino sketch publishes DHT11 health but does not publish dedicated PIR fault or LDR fault topics. Do not create PIR/LDR health indicators and label them as real diagnostics until those outputs are implemented.

5. HVAC scenarios and expected outputs

The Energy v3 sketch uses the following provisional thresholds. They are prototype assumptions for testing and have not yet been validated as campus comfort or energy-optimal limits.

Measurement	Current prototype rule
Temperature high	Temperature >= 31.0 °C
Humidity high	Relative humidity >= 75.0%
Environment HIGH	Temperature high OR humidity high
Environment NORMAL	Both temperature and humidity are below their high thresholds

Simulation preset values

Scenario payload	Effective temperature	Effective humidity	Environment result
NORMAL	25 °C	55%	NORMAL
HOT	33 °C	55%	HIGH
HUMID	25 °C	85%	HIGH
HOT_HUMID	33 °C	85%	HIGH

Crowd encoding

Payload	Meaning
0	NONE
1	LOW
2	MEDIUM
3	HIGH

Current detailed HVAC decision table

Environmental condition	Crowd = 0	Crowd = 1	Crowd = 2	Crowd = 3
NORMAL	OFF	LOW	LOW-MEDIUM	MEDIUM
High temperature + normal humidity	OFF	MEDIUM	HIGH	MEDIUM-HIGH
High humidity (temperature normal)	OFF	MEDIUM	HIGH	HIGH
HOT_HUMID (both high)	OFF	MEDIUM	HIGH	HIGH

Standard HVAC telemetry topic

The team-compatible 4-level output sitg7/room1/energy/hvac maps LOW-MEDIUM to MEDIUM and MEDIUM-HIGH to HIGH. The exact decision is preserved on sitg7/room1/energy/hvac_detail. Display the detailed topic as well if the presentation needs to show the original decision-table distinctions.

Testing order: Select SIMULATION first, then select the scenario, then the crowd level. Wait for the Arduino to receive the commands and check the reported effective readings and HVAC detailed setting. After Arduino restart, reselect all controls because the current firmware starts in LIVE / NORMAL / crowd 0.

6. Integration, restart and troubleshooting

MQTT topics and ownership

The group previously listed shared topics sitg7/room1/temp, sitg7/room1/hum, sitg7/room1/crowd and sitg7/room1/hvac for another board. The current Energy v3 sketch deliberately publishes under sitg7/room1/energy/... to avoid two devices publishing competing values to the same topic.

> Do not silently rename or merge topics. Before final integration, the group must agree which device owns the shared room-level topics and whether the dashboard will show energy-specific values, shared values, or both. The current Energy v3 sketch receives crowd simulation from sitg7/room1/cmd/crowd; it does not yet subscribe to the Crowd Management board's sitg7/room1/crowd topic.

Startup / reconnect sequence

- Turn on the phone hotspot or Wi-Fi network. The Arduino Uno WiFi Rev2 requires a 2.4 GHz-compatible Wi-Fi network.
- Power the Arduino and wait for Serial Monitor to report WiFi connected, MQTT connected and command wildcard subscription registered.
- Open IoT MQTT Panel, select the saved SIT Group 7 connection and connect if the cloud icon shows disconnected.
- Open Smart Campus Energy. New telemetry should arrive approximately every 2 seconds while the board is connected.
- Reselect LIVE/SIMULATION, scenario and crowd level after an Arduino restart. The current sketch does not restore those selections automatically.

Troubleshooting quick guide

Symptom	What to check
No dashboard messages	Phone connection is active; Text Output subscribes to the exact topic; dashboard prefix is disabled; Arduino MQTT is connected.
Commands do not change Arduino state	Combo Box topic and payload spelling; Arduino subscribed to sitg7/room1/cmd/#; select commands after board reconnects.
Scenario stays NORMAL	Check the received topic ends with /cmd/scenario and payload is exactly NORMAL, HOT, HUMID or HOT_HUMID.
Gauge stays blank or wrong	Topic is numeric; use effective temp/hum topic; check gauge min/max and payload parsing.
Phone or Arduino disconnects	Check hotspot/internet, ensure unique Client IDs, reconnect to public broker; public broker may occasionally be unavailable.
HVAC appears OFF	Crowd level 0 explicitly returns OFF. Set crowd 1/2/3 to test occupied HVAC scenarios.

Public broker and security

- HiveMQ public broker: broker.hivemq.com, port 1883, MQTT over TCP, no username/password, no TLS for this prototype setup.
- Use a unique Client ID for each connected device. Example IDs: sitg7-energy-uno02 (Energy Arduino), sitg7-dashboard-01 (dashboard device). Coordinate unique IDs with the other team boards.
- The public broker is not private. Do not publish personal information, credentials or sensitive data. Keep the agreed sitg7/ namespace.

Current limitations / next integration tasks

- Thresholds 31.0 °C and 75.0% RH remain provisional and need validation/justification for the report.
- HVAC is simulated status only; the prototype does not physically control building air-conditioning.
- Current MQTT sketch supports phone-driven simulation. For live group integration, update the Energy Arduino to subscribe to the Crowd Management board's agreed crowd topic and validate the 0-3 mapping.
- Google Sheets logging and historical energy-saving event records are future tasks; MQTT dashboard telemetry is not a durable historical database.